

Supplementary Material for: A Competitive Analysis of Online Multi-Agent Path Finding

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1 Proof of Theorem 1

Proof. We follow the reduction used in the proof of Theorem 3 in Ma et al. (2016) to construct an online MAPF instance that has a solution with makespan three if and only if a given $\leq 3,=3$ -SAT instance is satisfiable.

For each variable X_i in the $\leq 3,=3$ -SAT instance, we construct two “literal” agents, a_{iT} and a_{iF} , with start vertices s_{iT} and s_{iF} and goal vertices t_{iT} and t_{iF} , respectively. All literal agents have release time zero. For each literal agent, we construct two paths to get to its goal vertex in three time steps: a “shared” path, namely $\langle s_{iT}, u_{iT}, v_i, t_{iT} \rangle$ for a_{iT} and $\langle s_{iF}, u_{iF}, v_i, t_{iF} \rangle$ for a_{iF} , and a “private” path, namely $\langle s_{iT}, w_{iT}, x_{iT}, t_{iT} \rangle$ for a_{iT} and $\langle s_{iF}, w_{iF}, x_{iF}, t_{iF} \rangle$ for a_{iF} . The shared paths for a_{iT} and a_{iF} intersect at vertex v_i . Only one of the two paths can thus be used if a makespan of three is to be achieved. Sending literal agent a_{iT} (or a_{iF}) along the shared path corresponds to assigning *True* (or *False*) to X_i in the $\leq 3,=3$ -SAT problem instance.

For each clause C_j in the $\leq 3,=3$ -SAT instance, we construct a “clause” agent a_j with start vertex c_j , goal vertex d_j , and release time zero. It has multiple (but at most three) “clause” paths to get to its goal vertex in three time steps, which have a one-to-one correspondence to the literals in C_j . Every literal X_i (or $\bar{X}_i$) can appear in at most two clauses. If C_j is the first clause that it appears in, then the clause path is $\langle c_j, w_{iT}, b_j, d_j \rangle$ (or $\langle c_j, w_{iF}, b_j, d_j \rangle$). If C_j is the second clause that it appears in, a vertex α_j is introduced and the clause path is instead $\langle c_j, \alpha_j, x_{iT}, d_j \rangle$ (or $\langle c_j, \alpha_j, x_{iF}, d_j \rangle$). The clause path of each C_j with respect to any literal in that clause and the private path of the literal intersect. Only one of the two paths can thus be used if a makespan of three is to be achieved.

Figure 1 shows an example of the construction.

Suppose that a satisfying assignment to the $\leq 3,=3$ -SAT instance exists. Then, a solution with makespan three is obtained by sending literal agents of true literals along their shared paths, the other literal agents along their private paths, and clause agents along the clause paths corresponding to one of the true literals in those clauses.

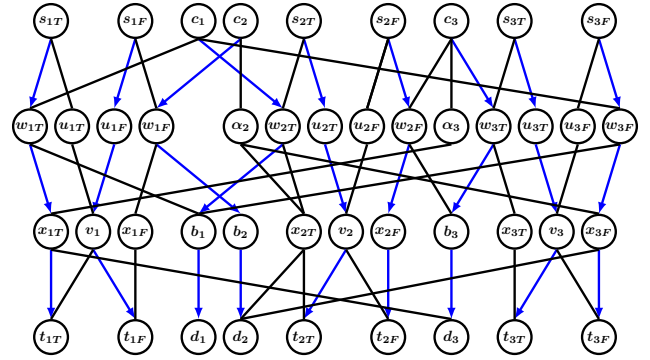


Figure 1: An online MAPF instance reduced from the $\leq 3,=3$ -SAT instance $(X_1 \vee X_2 \vee \bar{X}_3) \wedge (\bar{X}_1 \vee X_2 \vee \bar{X}_3) \wedge (X_1 \vee \bar{X}_2 \vee X_3)$. Clause C_1 is the first clause that literal X_1 appears in. The corresponding clause path is $\langle c_1, w_{1T}, b_1, d_1 \rangle$. Since clause C_2 is the second clause that X_2 appears in, vertex α_2 is introduced. The corresponding clause path is $\langle c_2, \alpha_2, x_{2T}, d_2 \rangle$. The blue (directed) edges represent one optimal solution to the online MAPF instance of makespan three, which corresponds to the satisfying assignment $(X_1, X_2, X_3) = (False, True, True)$.

Conversely, suppose that a solution with makespan three exists. Then, each clause agent traverses the clause path corresponding to one of the literals in that clause, and the corresponding literal agent traverses its shared path. Since the agents of a literal and its complement cannot both use their shared path if a makespan of three is to be achieved, we can assign *True* to every literal whose agent uses its shared path without assigning *True* to both the uncomplemented and complemented literals. If the agents of both literals use their private paths, we can assign *True* to any one of the literals and *False* to the other one. A solution to the online MAPF instance with makespan three thus yields a satisfying assignment to the $\leq 3,=3$ -SAT instance.

To summarize, the online MAPF instance has a solution with makespan three if and only if the $\leq 3,=3$ -SAT instance is satisfiable. Also, the online MAPF instance cannot have a solution with makespan smaller than three and always has

a solution with makespan four, even if the $\leq 3, = 3$ -SAT instance is unsatisfiable. For any $\epsilon > 0$, any approximation algorithm for online MAPF with ratio $4/3 - \epsilon$ thus computes a solution with makespan three whenever the $\leq 3, = 3$ -SAT instance is satisfiable and therefore solves $\leq 3, = 3$ -SAT. $\square$

References

- Ma, H. 2021. A Competitive Analysis of Online Multi-Agent Path Finding. In *ICAPS*.
- Ma, H.; Tovey, C.; Sharon, G.; Kumar, T. K. S.; and Koenig, S. 2016. Multi-Agent Path Finding with Payload Transfers and the Package-Exchange Robot-Routing Problem. In *AAAI*, 3166–3173.